

FIND Q SEARCH INDEX
VIEW CLUSTERING QUERY EXAMPLE

**i** M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.

|                                               | Database        | Collection | Index Name          | Status                                              | Queryable | Type         | Index Fields                                                                                             | Documents (Estimated)     |
|-----------------------------------------------|-----------------|------------|---------------------|-----------------------------------------------------|-----------|--------------|----------------------------------------------------------------------------------------------------------|---------------------------|
| SHORTCUTS<br>Cluster Metrics<br>Data Explorer | HR_ResumePortal | Resumes    | resume_bm25_index   | <b>READY</b><br><a href="#">View status details</a> |           | search       | "certifications",<br>"currentOrganisation",<br>"currentTitle",<br>"educationLevel",<br>"email" + 15 more | 146 (100%) indexed of 146 |
|                                               | HR_ResumePortal | Resumes    | resume_vector_index | <b>READY</b><br><a href="#">View status details</a> |           | vectorSearch | "embedding", "id",<br>"name", "email",<br>"phone" + 14 more                                              | 146 (100%) indexed of 146 |

[Learn more about Atlas Search](#)

## Phase 1

The screenshot displays the Postman application interface. On the left, a sidebar shows a collection of API requests under the name 'GenAI-Aug-HomeAssignment'. The main area shows a 'New Request' tab for the endpoint 'http://localhost:3000/v1/search/readiness'. The request method is 'GET'. Below the request details, the response is shown as a 200 OK status with a response time of 626 ms and a size of 463 B. The response body is a JSON object:

```
{
  "ready": true,
  "collection": "Resumes",
  "resumeCount": 146,
  "resumesWithEmbedding": 146,
  "embeddingModel": "mistral-embed",
  "embeddingDimension": 1024
}
```

On the right side of the interface, there is a 'New Chat' panel with a text input area and a 'Send' button. Below the input area, there are links for 'Diagnose localhost connection issue', 'Add status-specific test assertions', and 'Use a mock/monitor fallback'. At the bottom right, there is a 'Describe what you need. Press @ for context, / for skills.' prompt and an 'Auto' button.

0099...

File Edit Selection View Go Run Terminal Help

Explorer

Open Editors

02.Resume\_Retrieval\_Phasewise\_Implement...

HR.Resume\_Portal

dist

node\_modules

src

config

middleware

modules

Resumes\_TestData

shared

app.ts

server.ts

tests

uploads

.env

01.Resume\_Ingestion\_Phasewise\_Implement...

02.Resume\_Retrieval\_Phasewise\_Implement...

package-lock.json

package.json

tsconfig.json

Outline

Timeline

02.Resume\_Retrieval\_Phasewise\_Implementation 2.md

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Problems

Output

Debug Console

Terminal

Ports

node

1 Hidden ...

Update

Chat

Codex

Implement retrieval pipeline phases

Analyze my existing workspace and I have completed ingestion pipeline already an...

If no embeddings are available, the endpoint returns `503`:

```
{  "ready": false,  "reason": "No ingested resume e"}
```

The guide's sample says `resumes`, but ingestion actually writes to `Resumes`; the endpoint uses the existing collection name. Implementation is in

- `RetrievalValidationService.ts`,
- `retrievalRoutes.ts`, and
- `retrievalController.ts`.

Verification: TypeScript build passed; all 12 test suites and 36 tests passed.

I'll wait for your approval before proceeding to Phase 2.

+ 02.Resume\_Retrieval\_Phasewise\_Implementati...

Phase1 is passed. Proceed with phase2

+ 0 Agent Auto Balance

Local

Default permissions

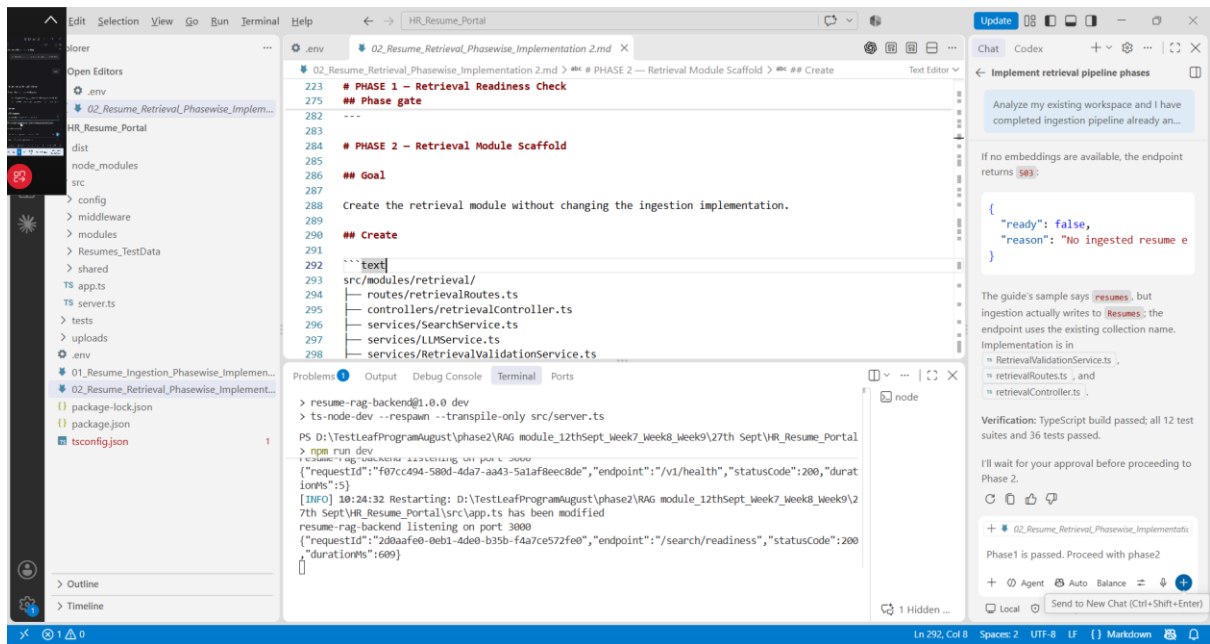
Ln 1, Col 1

Spaces: 2

UTF-8

LF

Markdown



## Phase2

```
02_Resume_Retrieval_Phasewise_Implementation 2.md
223 ## PHASE 1 - Retrieval Readiness Check
275 ## Phase gate
282 ---
283
284 ## PHASE 2 - Retrieval Module Scaffold
285
286 ## Goal
287
288 Create the retrieval module without changing the ingestion implementation.
289
290 ## Create
291
292 text
293 src/modules/retrieval/
294 |--- routes/retrievalRoutes.ts
295 |--- controllers/retrievalController.ts
296 |--- services/SearchService.ts
297 |--- services/LMService.ts
298 |--- services/RetrievalValidationService.ts
```

```
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
PS D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal
> npm run dev
[INFO] 10:24:32 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\app.ts has been modified
resume-rag-backend listening on port 3000
{"requestId":"2d0aaf0e-0eb1-4de0-b35b-f4a7ce572fe0","endpoint":"/search/readiness","statusCode":200,"durationMs":609}
```

There is no new Postman endpoint or request body in Phase 2. The existing readiness check remains `GET http://localhost:3000/v1/search/readiness` with no body. Its success response is:

```
{
  "ready": true,
  "collection": "Resumes",
  "resumeCount": 1,
  "resumesWithEmbedding": 1,
  "embeddingModel": "mistral-embed",
  "embeddingDimension": 1024
}
```

The TypeScript build and integration tests passed, including health and ingestion routes. Note: the guide mentions `/v1/resume/inject`, but this workspace exposes `/v1/resume/ingest`. I'll stop here and wait for your approval before Phase 3.

```
POST http://localhost:3000/v1/embeddings
Body
{
  "model": "mistral-embed",
  "dimension": 1024,
  "input": "senior agentic QA architect with RAG, DeepEval and MCP experience"
}
```

```
200 OK - 448 ms - 19.41 KB
Body
{
  "model": "mistral-embed",
  "dimension": 1024,
  "embedding": [
    -0.051055908, 0.3125,
    0.0167083749234375,
    0.0251617431640625,
    0.0244140625,
    0.021392822265625,
    0.022674569546875,
    0.028640747978125,
    -0.023501640625,
    -0.00835418791171875,
    ...
  ]
}
```

APITraining

Search

Invite Upgrade

Work locally with Git: Open your working directory in Postman to keep your API artifacts in sync with your codebase.

Try Native Git

COLLECTIONS

HealthCheckAPI agents

HR\_Resume\_Portal

Retrieval

Readiness

Phase3-Embeddings

HealthCheck

DBCheck

ResumeHealth

ResumeUpload

PDFTextExtraction

TextCleaning

SkillDictionary\_Detection

AlgorithmResumeParser

OptionalLLMParser

MistralResumeEmbedding

MongoDBResumeStorage

FullResumeIngestionService

ErrorHandling

LoggingrequestIds

ENVIRONMENTS

DOCUMENTS

SPECS

MOCKS

DATASETS

TOOLSETS

FLOWES

POST ErrorHandling

POST Loggingrequest

Retrieval

GET Readiness

POST Phase3-Embe

No Environment

AI

Send

Save

Share

Docs

Params

Authorization

Headers 10

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

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3

4

{

"model": "mistral-embed",

"input": "senior agentic QA architect with RAG, DeepEval and MCP experience"

}

Body

Cookies

Headers

Test Results

200 OK

448 ms

19.41 KB

Save Response

{}

JSON

Preview

Visualize

1018

-0.838670166015625,

1019

-0.858013916015625,

1020

0.04498291015625,

1021

0.838853271404375,

1022

0.05325173020125,

1023

-0.00954437255859375,

1024

0.84644775398625,

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0.0022945404052734375,

1026

0.002445220947265625,

1027

0.0036250697509765625,

1028

0.01406968994140625

1029

}

New Chat

Webex Session

Add assertions for embed

Create env vars for host

Parameterize input and add negative tests

Describe what you need. Press @ for context, / for skills.

Auto

Globals

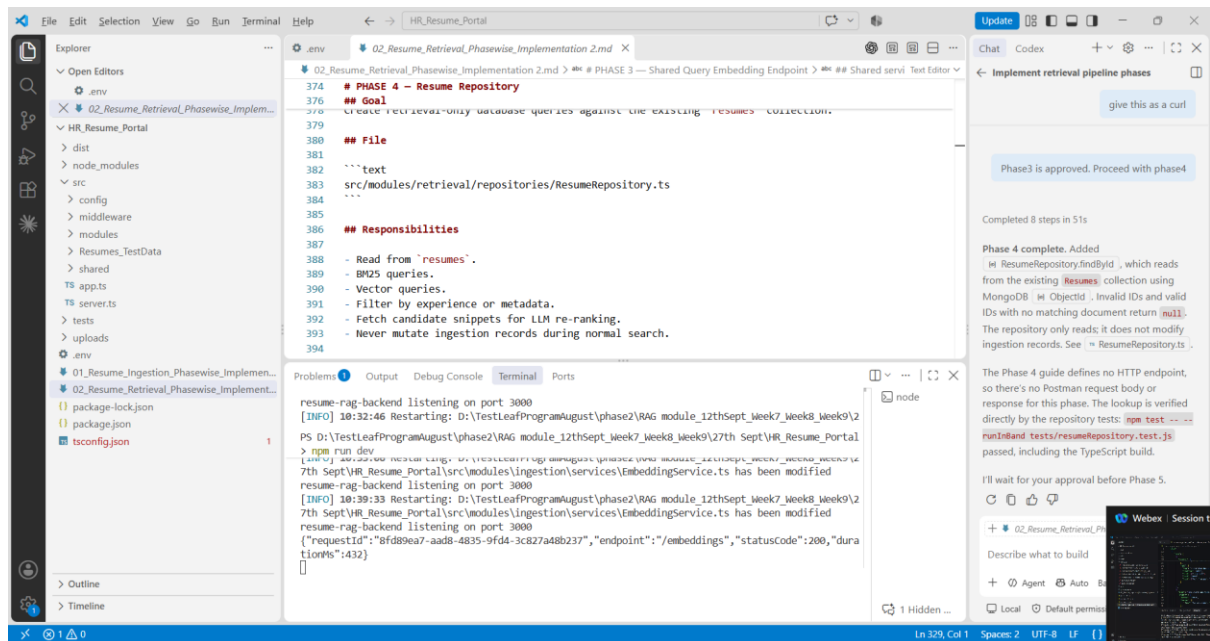
Vault

Tools

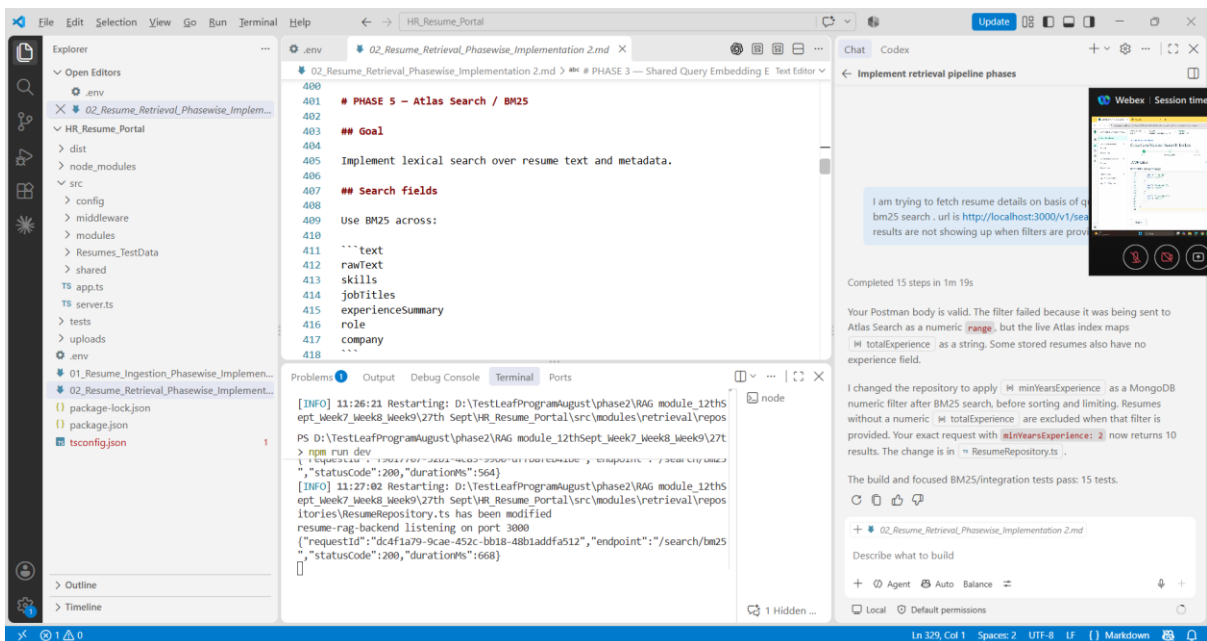
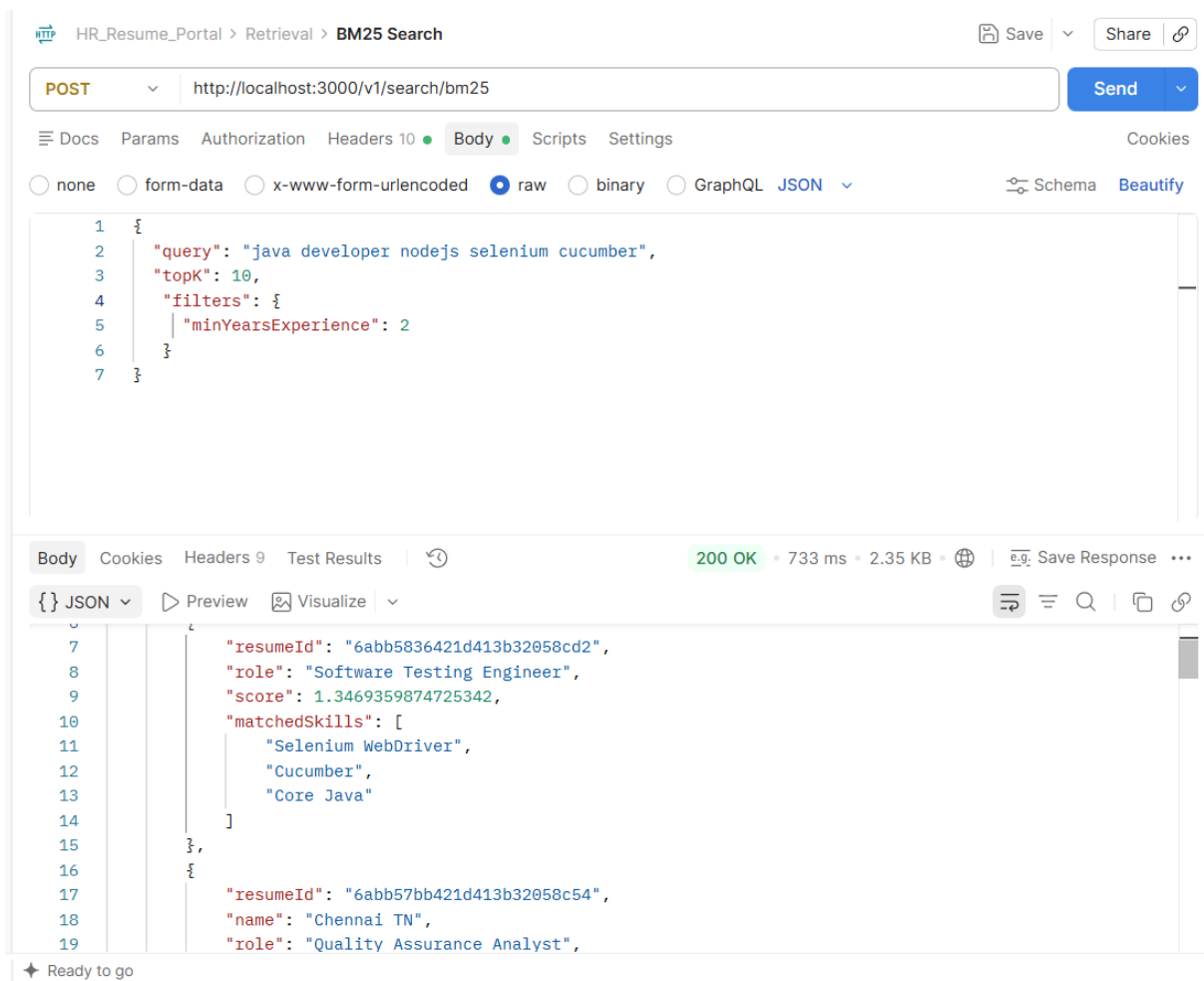
Ready to go

Phase3 is approvedd . Proceed with phase 4.

(There is not end point in phase 4)



Phase4 is approvedd . Proceed with phase 5.



Phase5 is approved . Proceed with phase 6

HR\_Resume\_Portal > Retrieval > VectorSearch

POST http://localhost:3000/v1/search/vector

Body

```
1 {
2   "query": "senior engineer who has built safe RAG and LLM evaluation systems",
3   "topK": 5
4 }
```

200 OK • 397 ms • 1.17 KB

Body

```
1 {
2   "mode": "vector",
3   "count": 5,
4   "results": [
5     {
6       "resumeId": "6abb5830421d413b32058ccc",
7       "role": "AI/ML Engineer with 2+ years of production experience architecting Autonomous Agents
8         and High-Throughput Data",
9       "vectorScore": 0.8397220373153687
10    },
11    {
12       "resumeId": "6abb5802421d413b32058c9d",
13       "role": "Performance Test Engineer with 4 years of experience in planning, designing, and
```

02\_Resume\_Retrieval\_Phasewise\_Implementation 2.md

PHASE 6 - MongoDB Vector Search

Goal

Use the query embedding to find semantically similar resumes.

Vector source

```
1 text
2 Query
3 ↓
4 Mistral EmbeddingService
5 ↓
6 1024-dimensional query vector
7 ↓
8 Atlas Vector Search
9 ↓
10 stored resume `embedding`
11
```

Terminal

```
PS D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9> npm run dev
...
[INFO] 11:34:49 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\route\s\retrievalRoutes.ts has been modified
resume-rag-backend listening on port 3000
```

Chat

Implement retrieval pipeline phases

Phase5 is approved . Proceed with phase 6

```
{
  "query": "senior engineer who has built safe RAG and
  "topK": 5
}
```

The live request returned five vector-ranked resumes. Example response:

```
{
  "mode": "vector",
  "count": 5,
  "results": [
    {
      "resumeId": "6abb5830421d413b32058ccc",
      "role": "AI/ML Engineer with 2+ years of producti
      "vectorScore": 0.8396419882774353
    }
  ]
}
```

Verification: Live Atlas/Mistral request succeeded; full build and test suite passed: 15 suites, 50 tests. I'll wait for your approval before Phase 7.

## Phase6 is approved. Proceed with phase7

The screenshot shows a VS Code editor with a file named `02_Resume_Retrieval_Phasewise_Implementation 2.md`. The code is divided into sections for Phase 6 and Phase 7.

```
476 # PHASE 6 - MongoDB Vector Search
477 ## Verify
478
479 # PHASE 7 - SearchService BM25 + Vector Methods
480 ## Goal
481
482 Introduce service-layer orchestration without hybrid behavior yet.
483
484 ## File
485
486 text
487 src/modules/retrieval/services/SearchService.ts
488
489 ## Methods
490
491 ts
492 bm25Search(query, filters, topK)
```

The chat interface on the right shows a message: "Phase5 is approved. Proceed with phase 6". Below it, a verification message states: "Verification: Live Atlas/Mistral request succeeded; full build and test suite passed: 15 suites, 50 tests. I'll wait for your approval before Phase 7." Another message says: "Phase6 is approved. Proceed with phase7".

The terminal at the bottom shows the command `npm run dev` and the output of the application, including a restart message and a log entry: `[INFO] 11:34:49 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\route s\retrievalRoutes.ts has been modified resume-rag-backend listening on port 3000`.

The screenshot shows a VS Code editor with a file named `02_Resume_Retrieval_Phasewise_Implementation 2.md`. The code is divided into sections for Phase 7 and Phase 8.

```
565 # PHASE 7 - SearchService BM25 + Vector Methods
566 ## Methods
567
568
569 ## Verify
570
571 Each method can be tested independently and returns a normalized candidate
572 shape.
573
574 Recommended normalized candidate:
575
576
577 ts
578 interface SearchCandidate {
579   resumeId: string;
580   name?: string;
581   role?: string;
582   company?: string;
583   skills?: string[];
584 }
```

The chat interface on the right shows a message: "Phase6 is approved. Proceed with phase7". Below it, a verification message states: "Verification: Live Atlas/Mistral request succeeded; full build and test suite passed: 15 suites, 52 tests. I'll wait for your approval before Phase 8." Another message says: "Phase7 adds no endpoint, so there's no new Postman request body. The service methods return candidates like: { 'resumeId': '691db80aa895776f97b6eca6', 'name': 'Ava Example', 'role': 'QA Engineer', 'company': 'Example Ltd', 'skills': ['Java', 'Selenium'], 'bm25Score': 4.2, 'sources': ['bm25'] }".

The terminal at the bottom shows the command `npm run dev` and the output of the application, including a restart message and a log entry: `[INFO] 11:37:28 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\utils\candidateUpper.ts has been modified resume-rag-backend listening on port 3000`.

There is no endpoint for phase7

Phase7 is approved . Proceed with phase8

```
607 # PHASE 8 - Hybrid Search
608 ## Goal
609
610 Run lexical and semantic retrieval independently and in parallel.
611
612 ## Method
613
614
615
616 hybridSearch(query, filters, options)
617
618
619 ## Required flow
620
621
622
623
624
625
626
```

Problems Output Debug Console Terminal Ports

```
resume-rag-backend listening on port 3000
[INFO] 11:42:24 Restarting: D:\TestLeafProgram\phase2\RAG module_12ths
PS D:\TestLeafProgram\phase2\RAG module_12th\Sept_Week8_Week9\27th
> npm run dev
[INFO] 11:43:47 Restarting: D:\TestLeafProgram\phase2\RAG module_12ths
pt_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\services\SearchService.ts has been modified
resume-rag-backend listening on port 3000
{"requestId":"53aa92c1-67fe-4f67-bda9-ed6ac9f47ed4","endpoint":"/search/hybrid","bm25Ms":582,"embeddingMs":472,"vectorMs":322,"statusCode":200,"durationMs":842}
```

Chat Codex

Expand window to show video

Phase7 is approved . Proceed with phase8

Phase 8 complete: Added `POST /v1/search/hybrid`. It runs BM25 and vector retrieval concurrently, keeps their scores separate, and returns/logs component timings. It does not merge candidates, combine scores, or rerank.

Postman request

POST `http://localhost:3000/v1/search/hybrid`  
Content-Type: `application/json`

```
{
  "query": "agentic QA architect with RAG evaluation",
  "topK": 5,
  "filters": {
    "minYearsExperience": 2
  }
}
```

A live request returned the same resume in both independent result lists:

```
{
  // ...
}
```

> 8 files changed +235 -4

Keep Undo

+ 02\_Resume\_Retrieval\_Phasewise\_Implementation 2.md

Describe what to build

+ Agent Auto Balance

Local Default permissions

GET Readiness POST Phase3-Embe POST VectorSearch POST BM25 Search POST HybridSearch No Environment

HR\_Resume\_Portal > Retrieval > HybridSearch

POST http://localhost:3000/v1/search/hybrid

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "query": "agentic QA architect with RAG evaluation",
3   "topK": 5,
4   "filters": {
5     "minYearsExperience": 2
6   }
7 }
```

Body Cookies Headers 9 Test Results 200 OK • 849 ms • 860 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "mode": "hybrid-debug",
3   "bm25": [
4     {
5       "resumeId": "6abb5830421d413b32058ccc",
6       "score": 0.7815473079681396
7     }
8   ],
9   "vector": [
10    {
11      "resumeId": "6abb5830421d413b32058ccc",
12      "score": 0.853652834892273
13    }
14  ]
15 }
```

## Phase 8 is approved . Proceed with phase 9

The screenshot shows a VS Code editor with a file named `02_Resume_Retrieval_Phasewise_Implementation 2.md`. The code is in a text editor with line numbers 690 to 708. It contains a section for Phase 9: Merge + Deduplicate Candidates, with a goal to create a candidate pool for LLM re-ranking. A utility function `deduplicate` is defined, and a deduplication key `resumeId` is specified.

Below the code editor is a terminal window showing the command `npm run dev` and the output of the application, including a message about restarting the server and a JSON response from the `/search/hybrid` endpoint.

On the right side, there is a chat window with a video call icon. The chat contains a message from the AI: "Phase7 is approved . Proceed with phase8". Below this, there is a code block showing a JSON object with timing information: `{ "timings": { "bm25Ms": 582, "embeddingMs": 472, "vectorMs": 322 } }`. Another message from the AI says: "Implementation is in SearchService.ts, with the route in retrievalRoutes.ts. Live search succeeded; the full build and test suite passed: 16 suites, 55 tests. I'll wait for your approval before Phase 9." A third message says: "Phase8 is approved. Proceed with phase9".

The screenshot shows a REST client interface for the `HR_Resume_Portal` project, specifically for the `Retrieval > Deduplication` endpoint. The request is a `POST` to `http://localhost:3000/v1/search/hybrid`. The request body is a JSON object:

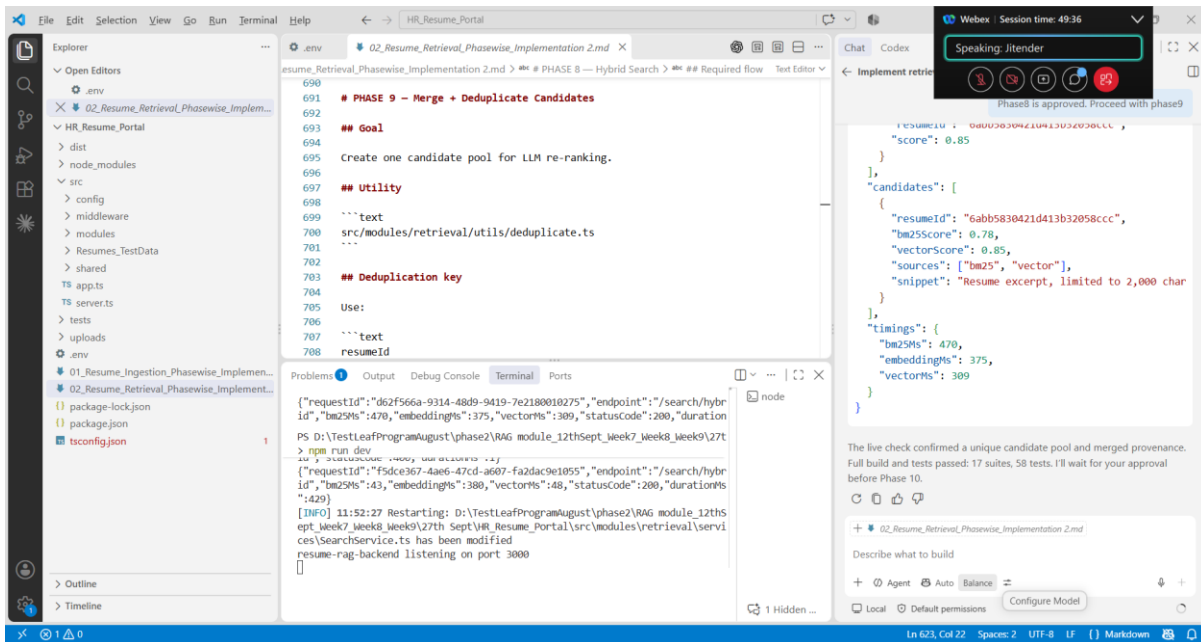
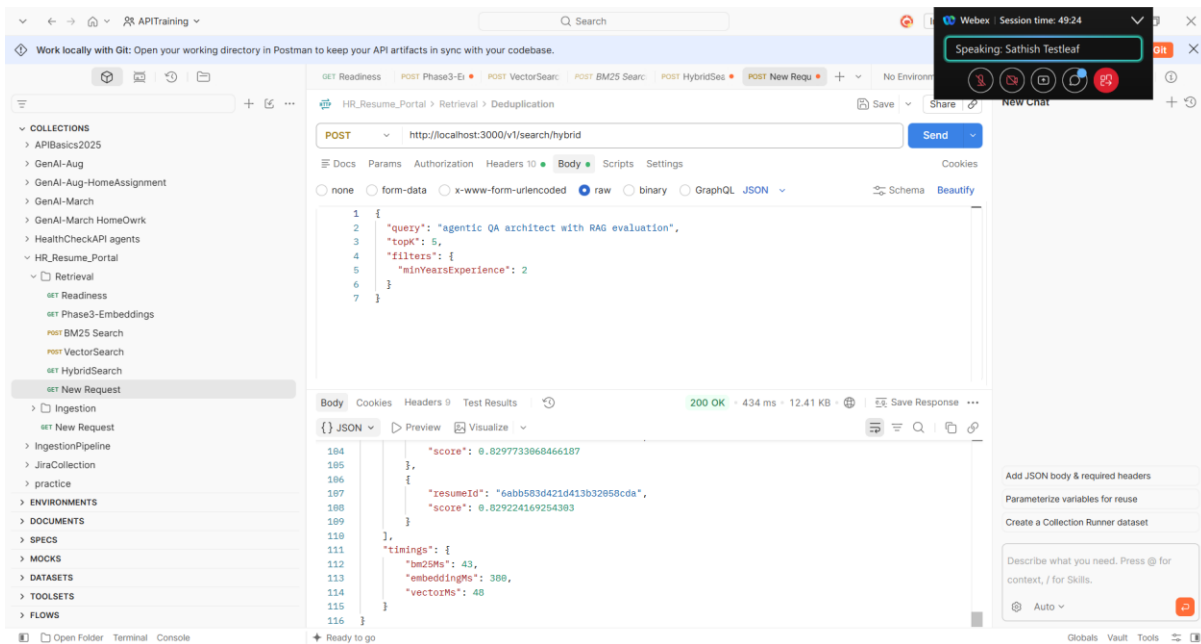
```
1 {
2   "query": "agentic QA architect with RAG evaluation",
3   "topK": 5,
4   "filters": {
5     "minYearsExperience": 2
6   }
7 }
```

The response is a `200 OK` status with a response time of 434 ms and a size of 12.41 KB. The response body is a JSON object:

```
1 {
2   "mode": "hybrid-debug",
3   "candidates": [
4     {
5       "resumeId": "6abb5830421d413b32058ccc",
6       "role": "AI/ML Engineer with 2+ years of production experience architecting Autonomous Agents and High-Throughput Data",
7       "skills": [
8         "GitHub",
9         "RAG",
10        "GenAI",
11        "Langchain",
12        "Langgraph",

```

At the bottom of the interface, there is a status bar that says "Ready to go".



## Phase 9 is approved . Proceed with Phase 10

The screenshot displays a development environment with a code editor, a terminal, and a chat interface.

**Code Editor:** The file `02_Resume_Retrieval_Phasewise_Implementation 2.md` is open. It contains the following code:

```
746 # PHASE 10 - Groq LLM Service
747
748 ## Goal
749
750 Implement LLM re-ranking, summarization, and metadata support.
751
752 ## File
753
754 ```text
755 src/modules/retrieval/services/LLMService.ts
756 ```
757
758 ## Model
759
760 ```text
761 meta-llama/llama-4-scout-17b-16e-instruct
762 ```
763
```

**Terminal:** The terminal shows the following output:

```
[INFO] 11:52:27 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\services\LLMService.ts has been modified
PS D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\services\LLMService.ts has been modified
> npm run dev
[INFO] 11:55:56 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\services\LLMService.ts has been modified
resume-rag-backend listening on port 3000
[INFO] 11:55:56 Restarting: D:\TestLeafProgramAugust\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal\src\modules\retrieval\services\LLMService.ts has been modified
resume-rag-backend listening on port 3000
```

**Chat Interface:** The chat interface shows a conversation with Jitender. The messages are:

- Phase8 is approved. Proceed with phase9
- Phase 9 is approved . Proceed with Phase 10

The chat interface also shows a progress bar indicating that 15 steps were completed in 2m 21s.

**Phase 10 complete.** Added the Groq-backed `LLMService` with `rerankCandidates`, `summarizeCandidateFit`, and `extractMetadata`. Reranking uses structured JSON, validates every returned ID against the supplied candidates, and rejects invented or duplicate IDs. Summaries are grounded in the supplied candidate data; metadata uses the existing ingestion schema and is not persisted.

**Phase 10 defines service methods, not HTTP routes, so there's no Postman request yet.** The service returns ranked results with a model name, summaries with a `resumeld`, and validated parsed metadata, respectively. The Postman endpoints are introduced in later phases.

**Verification:** Full build and test suite passed: 18 suites, 65 tests. Groq calls were tested with an injected mock client, not against the live Groq API. I'll wait for your approval before Phase 11.

The chat interface also shows a list of files: `02_Resume_Retrieval_Phasewise_Implementation 2.md`.

## Phase 10 is approved . Proceed with Phase 11

GET Readiness POST Phase3-Embedding POST VectorSearch POST HybridSearch POST New Request No Environment

HR\_Resume\_Portal > Retrieval > New Request

POST http://localhost:3000/v1/search/rerank

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "query": "Need a senior QA architect experienced in RAG, DeepEval, MCP and enterprise GenAI governance",
3   "candidates": [
4     {
5       "resumeId": "691db80aa895776f97b6eca6",
6       "snippet": "Test Architect with 13+ years, RAG, DeepEval, MCP, Agentic QA System, LLM Evaluation..."
8     },
9   ],
10  "topK": 10
11 }
```

Body Cookies Headers 9 Test Results 200 OK • 1.12 s • 711 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "results": [
3     {
4       "resumeId": "691db80aa895776f97b6eca6",
5       "rank": 1,
6       "relevanceScore": 0.95,
7       "reason": "Senior Test Architect with 13+ years experience explicitly lists RAG, DeepEval, MCP, and LLM evaluation, aligning closely with the required QA architect skills; only enterprise GenAI governance is not directly mentioned, but the overall profile strongly matches."
8     },
9   ],
10  "model": "openai/gpt-oss-120b"
11 }
```

File Edit Selection View Go Run Terminal Help

HR\_Resume\_Portal

02\_Resume\_Retrieval\_Phasewise\_Implementation 2.md

```
801 # PHASE 11 - LLM Re-Ranking Endpoint
802
803 ## Endpoint
804
805 http
806 POST /v1/search/rerank
807
808 ## Learner request
809
810 json
811 {
812   "query": "Need a senior QA architect experienced in RAG, DeepEval, MCP and enterprise GenAI governance",
813   "candidates": [
814     {
815       "resumeId": "691db80aa895776f97b6eca6",
816       "snippet": "Test Architect with 13+ years, RAG, DeepEval, MCP, Agentic QA System, LLM Evaluation..."
817     }
818   ],
819   "topK": 10
820 }
821
```

Problems Output Debug Console Terminal Ports

```
> ts-node-dev --respawn --transpile-only src/server.ts
PS D:\TestLeafProgram\August\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal> npm run dev
resume-rag-backend listening on port 3000
{"requestId":"663a11ea-7a6d-4e14-90b6-00eb1fd9570d","endpoint":"/search/rerank","statusCode":502,"durationMs":203}
{"requestId":"dba0f0d9-5a0a-4c93-b019-709b1213b7ef","endpoint":"/search/rerank","statusCode":502,"durationMs":119}
{"requestId":"3620e79c-0e05-48ed-802f-5edcd20b75c","endpoint":"/search/rerank","statusCode":502,"durationMs":81}
PS D:\TestLeafProgram\August\phase2\RAG module_12thSept_Week7_Week8_Week9\27th Sept\HR_Resume_Portal>
```

Webex 03:20:42

Speaking: Prabakaran Testleaf

Implement retrieval

I am trying to perform reranking but it is giving me 502 error

Completed 17 steps in 1m 37s

The 502 was caused by a duplicated prefix in .env :

GROQ\_MODEL=GROQ\_MODEL=openai/gpt-oss-120b

I corrected it to:

GROQ\_MODEL=openai/gpt-oss-120b

The server was still using its old environment, so I restarted it. Your same rerank request now returns 200 with a ranked result and "model": "openai/gpt-oss-120b". The refreshed API is running on port 3000.

+ 02\_Resume\_Retrieval\_Phasewise\_Implementation 2.md

Describe what to build

+ Local Agent Auto Balance

Local Default permissions

Ln 701, Col 4 Spans: 2 UTF-8 LF Markdown

Phase 11 is approved . Proceed with phase 12

The image displays two screenshots from a development environment. The top screenshot shows the Postman interface with a REST client request for the endpoint `http://localhost:3000/v1/search/summarize`. The request body is a JSON object:

```
{  "query": "Senior QA architect with GenAI RAG and evaluation experience",  "candidate": {    "resumeId": "691db80aa895776f97b6eca6",    "snippet": "13+ years of experience in QA architecture, RAG, DeepEval, MCP, and agentic QA systems."  },  "style": "short",  "maxTokens": 150}
```

The response is a 200 OK status with a JSON body:

```
{  "resumeId": "691db80aa895776f97b6eca6",  "summary": "The candidate's 13+ years designing QA architectures, including GenAI-based RAG and evaluation tools such as DeepEval, directly match the senior QA architect role."}
```

The bottom screenshot shows the VS Code editor with a file named `02_Resume_Retrieval_Phasewise_Implementation 2.md`. The code defines Phase 11 as the LLM Re-Ranking Endpoint and Phase 12 as the Candidate Summarization endpoint. The goal for Phase 12 is to generate optional candidate-fit summaries. The endpoint is a POST request to `/v1/search/summarize` with a JSON body:

```
{  "query": "Senior QA architect with GenAI RAG and evaluation experience"}
```

The terminal output shows the command `ts-node-dev --respawn --transpile-only src/server.ts` being executed, and the chat window displays the response from the live request:

```
{  "resumeId": "691db80aa895776f97b6eca6",  "summary": "The candidate's 13+ years designing QA ar"}
```

Phase 12 is approved . Proceed with phase 13

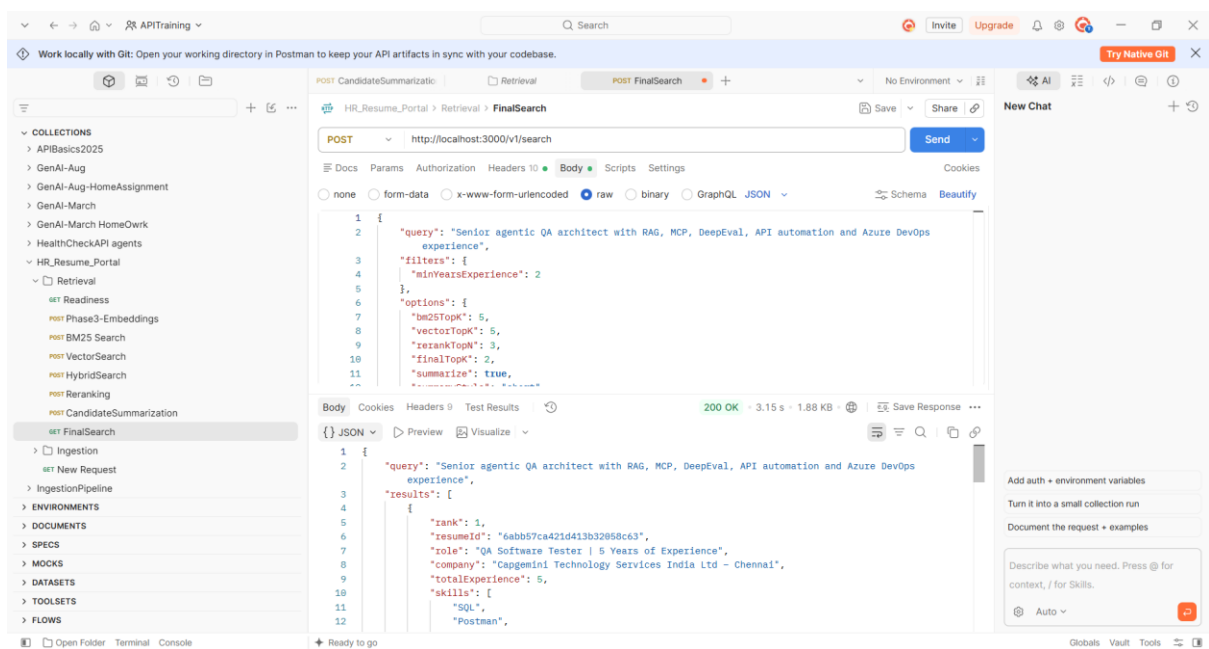
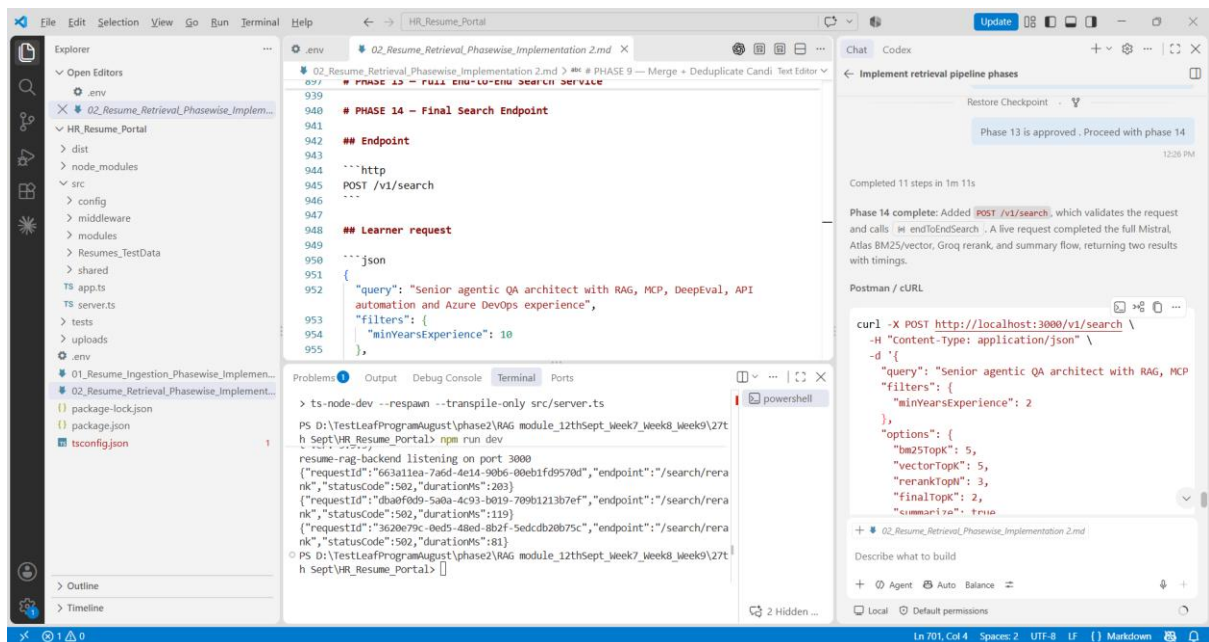
The screenshot displays the Visual Studio Code (VS Code) interface during the development of a search service. The main editor shows the file `02_Resume_Retrieval_Phasewise_Implementation 2.md` with the following code:

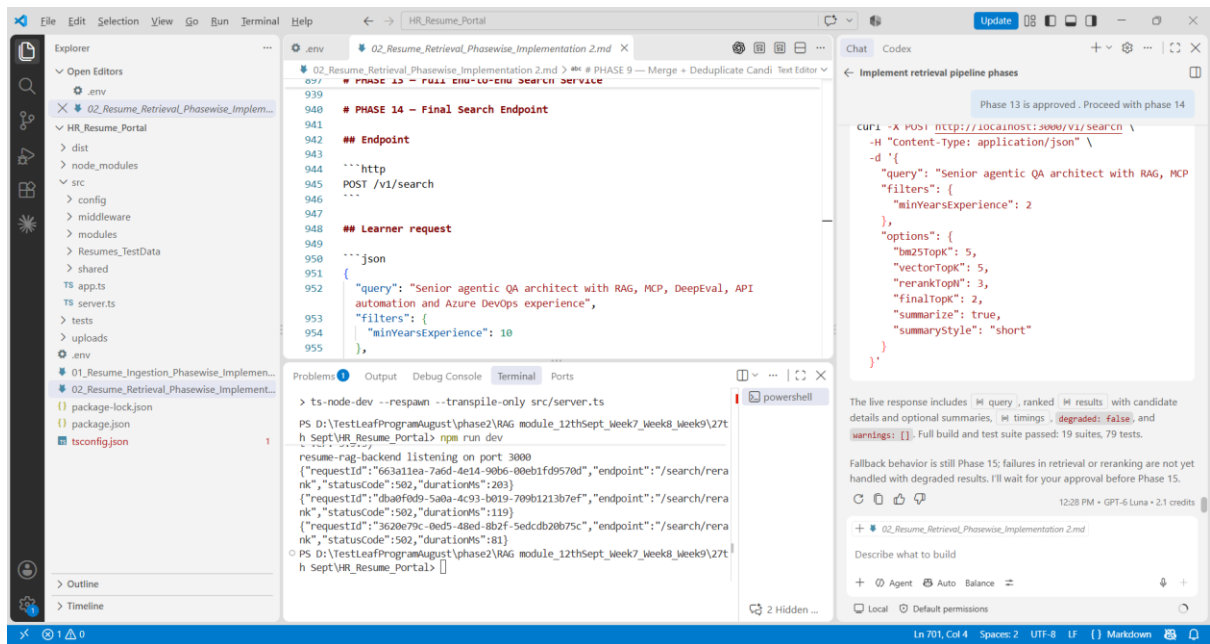
```
897 ## PHASE 13 - Full End-to-End Search Service
898
899 ## Goal
900
901 Build the final synchronous search pipeline.
902
903 ## Method
904
905 ```ts
906 endToEndSearch(query, filters, options)
907 ```
908
909 ## Final flow
910
911 ```text
912 1. Validate request
913 2. Generate query embedding
914 3. Run BM25 search
```

The terminal window shows the command `ts-node-dev --respawn --transpile-only src/server.ts` being executed, resulting in the server starting and listening on port 3000. It also displays several JSON log entries for search requests.

The chat window on the right, titled "Implement retrieval pipeline phases", provides a summary of the progress: "Phase 11 is approved. Proceed with phase 12" and "Phase 12 is approved. Proceed with phase 13". It also includes a "Restore Checkpoint" button and a completion status: "Completed 20 steps in 2m 31s". The chat further details the implementation of Phase 13, noting that it adds no endpoint and is ready for the `/v1/search` endpoint in Phase 14. Verification steps are listed: "Full build and test suite passed: 19 suites, 77 tests. I'll wait for your approval before Phase 14."

Phase 13 is approved . Proceed with phase 14





## Phase 14 is approved . Proceed with phase 15

The screenshot displays a development environment with three main components:

- Code Editor:** Shows the implementation of Phase 15 - Fallback Logic. The code includes comments for the goal, required fallbacks, and LLM re-rank failure handling. It details the fallback ordering: BM25 priority, then vector.
- Terminal:** Displays the command `ts-node-dev --respawn --transpile-only src/server.ts` and its output, showing the server starting on port 3000 and logging request details for the `/search/rerank` endpoint.
- Chat Window:** Contains messages indicating that Phase 13 and Phase 14 are approved, and Phase 15 is complete. It also includes a verification message stating that the full suite passed 19 suites and 84 tests.

Phase 15 is approved . Proceed with phase 16

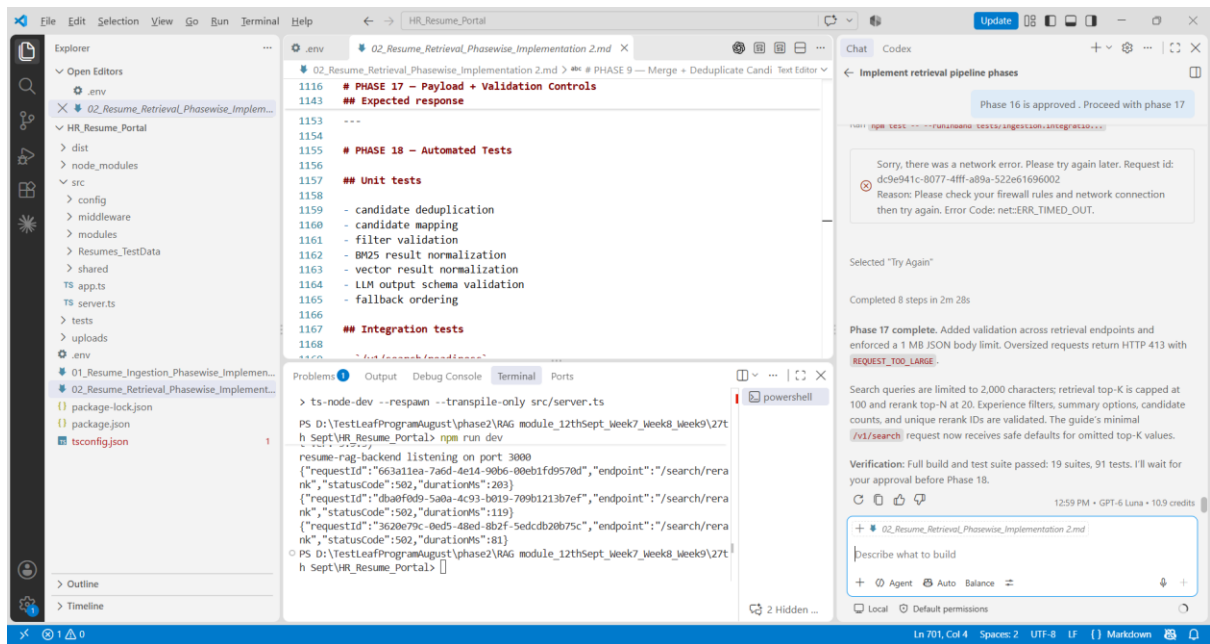
The screenshot displays a development environment with three main panels:

- Explorer:** Shows the project structure with folders like `dist`, `node_modules`, `src`, `config`, `middleware`, `modules`, `Resumes_TestData`, `shared`, `app.ts`, `server.ts`, `tests`, `uploads`, and `env`.
- Editor:** Displays the file `02_Resume_Retrieval_Phasewise_Implementation 2.md`. The code defines a `PHASE 16 -- Logging + Timings` section, including a `Required log shape` with fields like `requestId`, `endpoint`, `method`, `durationMs`, `statusCode`, and `componentTimings`. A `Verify` section is also present.
- Terminal:** Shows the command `ts-node-dev --respawn --transpile-only src/server.ts` being executed. The output indicates that the resume-rag-backend is listening on port 3000 and provides sample log entries for the `/search/rerank` endpoint.
- Chat:** A sidebar chat window titled "Implement retrieval pipeline phases" shows a sequence of status updates: "Phase 14 is approved . Proceed with phase 15", "Phase 15 is approved . Proceed with phase 16", and "Phase 16 complete. Logs now include the request ID, full /v1/... endpoint, HTTP method, status, total duration, and nested componentTimings...". It also mentions a full build and test suite passing with 19 suites and 85 tests.

Phase 16 is approved . Proceed with phase 17

The screenshot displays a development environment with three main components:

- Editor:** A TypeScript file named `02_Ressume_Retrieval_Phasewise_Implementation 2.md` is open. It contains code for Phase 17, including a comment `## PHASE 17 -- Payload + Validation Controls`, an `Expected response` block, and a list of unit tests for candidate deduplication, mapping, filter validation, BM25 normalization, vector normalization, LLM output schema validation, and fallback ordering. Integration tests are also listed.
- Terminal:** The terminal shows the execution of `ts-node-dev --respawn --transpile-only src/server.ts`. The output indicates that the resume-rag-backend is listening on port 3000 and provides details for three test requests, including their request IDs, status codes, and durations.
- Chat:** A chat window titled "Implement retrieval pipeline phases" shows a message stating "Phase 16 is approved . Proceed with phase 17". Below this, a network error is detailed: "Sorry, there was a network error. Please try again later. Request id: d59e941c-8077-48ff-a89a-522e61696002. Reason: Please check your firewall rules and network connection then try again. Error Code: net:ERR\_TIMED\_OUT." The chat also shows a "Try Again" button and a completion status of "Completed 8 steps in 2m 28s".



Phase 17 is approved . Proceed with phase 18

The screenshot shows the VS Code interface with the file explorer on the left, the main editor in the center, and the chat sidebar on the right. The main editor displays the file `02_Resume_Retrieval_Phasewise_Implementation 2.md` with the following content:

```
1155 # PHASE 18 -- Automated Tests
1156 ## Unit tests
1157 - candidate deduplication
1158 - candidate mapping
1159 - filter validation
1160 - BM25 result normalization
1161 - vector result normalization
1162 - LLM output schema validation
1163 - fallback ordering
1164
1165 ## Integration tests
1166 - /v1/search/readiness
1167 - /v1/embeddings
1168 - /v1/search/bm25
1169 - /v1/search/vector
1170 - /v1/search/hybrid
1171 - /v1/search/rerank
```

The terminal at the bottom shows the command `ts-node-dev --respawn --transpile-only src/server.ts` being executed, and the output indicates that the application is running on port 3000.

The chat sidebar on the right shows a message from the AI assistant: "Phase 17 is approved. Phase 18 is primarily verification, and the workspace already has unit and integration coverage from the prior phases; I'll compare those tests against the guide's checklist and add only any missing E2E coverage rather than duplicating existing tests." Below this message, there is a button that says "Phase 17 is approved . Proceed with phase 18".

The screenshot shows the VS Code interface with the file explorer on the left, the main editor in the center, and the chat sidebar on the right. The main editor displays the file `02_Resume_Retrieval_Phasewise_Implementation 2.md` with the following content:

```
1338 # 12. Final Rule for Learners
1339 Complete every search component
1340 Verify fallback + end-to-end search
1341 Application complete
1342 The implementation objective is not simply to create endpoints. The learner
1343 must prove each phase with its request, expected response, MongoDB result,
1344 and test checkpoint before moving forward.
```

The terminal at the bottom shows the command `ts-node-dev --respawn --transpile-only src/server.ts` being executed, and the output indicates that the application is running on port 3000.

The chat sidebar on the right shows a message from the AI assistant: "Phase 17 is approved. Phase 18 is primarily verification, and the workspace already has unit and integration coverage from the prior phases; I'll compare those tests against the guide's checklist and add only any missing E2E coverage rather than duplicating existing tests." Below this message, there is a button that says "Phase 17 is approved . Proceed with phase 18".

